

DUAL MODE SEWER FLOWMETER

E-Flow 3

Four sensor area velocity flowmeter for sewer maintenance holes. Dual radar in normal flow, dual ultrasonic and sealed pressure in surcharge.

E-Flow 3 is built for harsh, demanding sewer environments. A pair of contactless radars at the pipe obvert measures velocity and depth in normal flow, so nothing sits in the flow. When the sewer surcharges and the free surface is lost, a dual 1 MHz ultrasonic Doppler velocity sensor and a sealed 316 stainless steel pressure transducer take over seamlessly. One sensor body, one record, from dry weather flow through to full surcharge.



4 sensors	0.05 to 20 m/s	0 to 10 m	150 mA	RS485
ONE BODY	VELOCITY RANGE	DEPTH RANGE	AT 12 VDC	MODBUS RTU

NORMAL FLOW · CONTACTLESS

Dual radar

A 24 GHz Doppler radar reads surface velocity and a 120 GHz FMCW radar reads depth, both from above the flow. No wetted parts, so grease, rag and silt cannot foul the measurement.

SURCHARGE · CONTACT

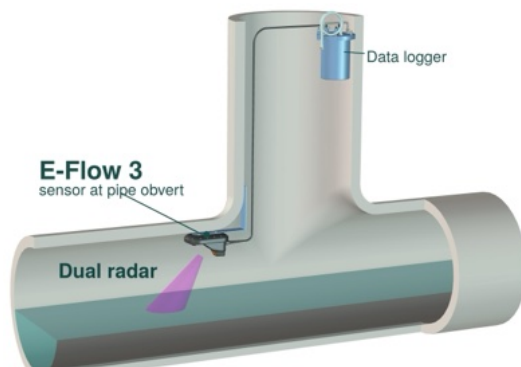
Dual ultrasonic and pressure

Dual 1 MHz ultrasonic Doppler transducers measure velocity through the water column, and a sealed 316 stainless steel high accuracy pressure transducer measures depth to 10 m.

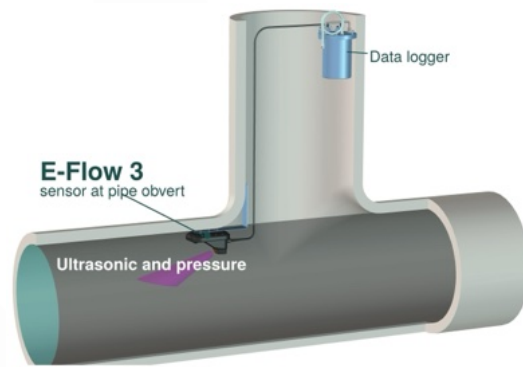
ALL CONDITIONS

Seamless changeover

The sensor selects the appropriate pair automatically, with no blind zone and no gap in the record. Composite velocity, depth and flow are output alongside each sensor's own value and status.



Normal flow. Dual radar measures velocity and depth from the obvert; the sensor never touches the flow.

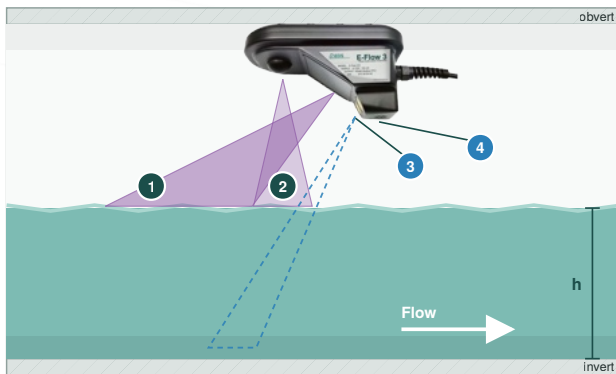


Surcharged sewer. Ultrasonic Doppler and the sealed pressure transducer keep measuring when the pipe runs full.

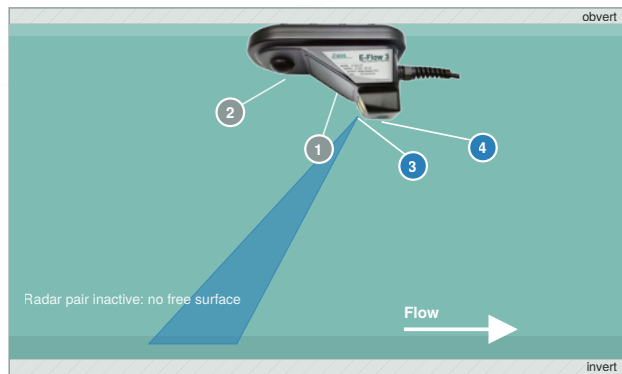
01 Principle of operation

E-Flow 3 is an area velocity flowmeter. Flow rate is computed in the sensor from mean velocity and the wetted area derived from depth and the configured pipe profile. Four sensors share one housing at the pipe obvert and are used in two pairs according to the fill state of the pipe, so a valid velocity and depth are available from low flow through to full surcharge.

Normal flow, partially full



Surcharged, pipe full



- 1 Velocity radar, 24 GHz 2 Depth radar, 120 GHz 3 Dual ultrasonic velocity, 1 MHz 4 Pressure transducer, 316 stainless

Figure 1. Measurement arrangement in partially full and surcharged conditions. The radar pair reads the free surface from above the flow; the ultrasonic pair and pressure transducer measure through the water column when the pipe is full.

Normal flow: dual radar, contactless

The 24 GHz continuous wave Doppler radar illuminates the surface upstream of the sensor and measures the Doppler shift of the return to obtain surface velocity, corrected to mean channel velocity. The 120 GHz FMCW radar measures the distance to the free surface to give depth. Both transducers are above the flow and never contact the flow, so rag, grease and silt cannot affect them.

Surcharge: dual ultrasonic and sealed pressure

Dual 1 MHz ultrasonic Doppler transducers at the sensor face transmit into the water column and measure the Doppler shift from particles carried with the flow, giving velocity when there is no free surface. A sealed 316 stainless steel high accuracy pressure transducer measures the head of water above the sensor face, compensated for dynamic pressure, and reads to 10 m of surcharge. On the bidirectional model, ultrasonic velocity is signed and reverse flow is reported.

Seamless changeover

The sensor monitors all four elements and selects the appropriate pair as the pipe fills and drains. The ultrasonic and pressure sensors remain available and cross checked while the radar pair is primary, so the transition into and out of surcharge is continuous, with no blind zone.

Outputs

Composite velocity, depth and flow rate are the primary outputs. Each sensor's own value and a status word are also available over Modbus, so the active pair and the health of every element can be monitored remotely.

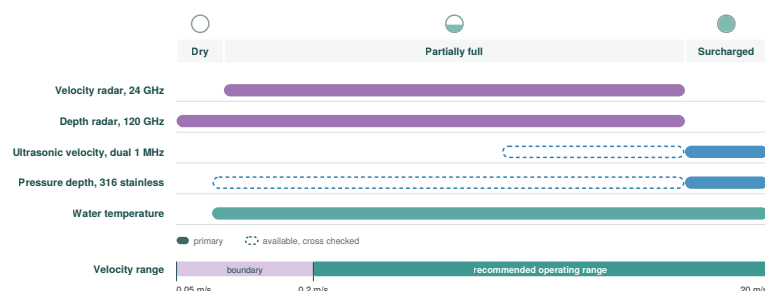


Figure 2. Sensor activity against pipe fill state, and velocity operating range.

02 Sensor anatomy and measurement geometry

The depth radar is housed in the dome at the upstream end of the body and looks straight down at the free surface. The velocity radar is behind the sloping face ahead of the nose block, aimed upstream along the approaching surface. The dual ultrasonic transducers are on the angled front face of the nose block and the pressure transducer port is on its underside, so these are the only elements in contact with the flow, and only when the pipe is full. Depth is converted to wetted area using the configured pipe profile, and flow is the product of wetted area and mean velocity.

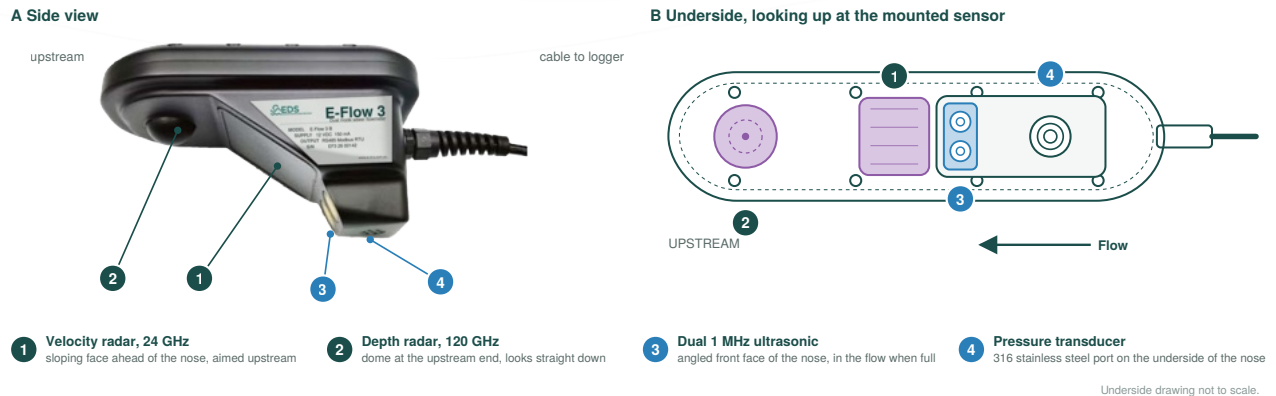


Figure 3. Sensor anatomy. Side view with element positions, and the underside as seen from the invert.

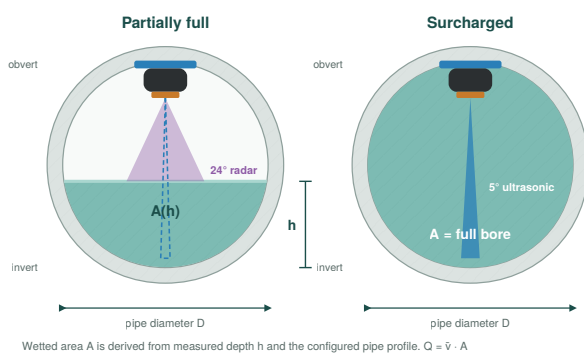


Figure 4. Pipe cross section. Wetted area A(h) from measured depth; radar footprint in normal flow, ultrasonic beam in surcharge.

Radars geometry

The velocity radar has a $24^\circ \times 12^\circ$ antenna pattern and is aimed upstream so that the footprint sits on the approaching surface. The depth radar has a $14^\circ \times 10^\circ$ pattern and looks vertically at the surface beneath the sensor. Both need a free surface with enough texture to return a signal, which is the normal case in a flowing sewer.

Ultrasonic and pressure geometry

The dual 1 MHz ultrasonic transducers project a narrow 5° beam into the water column and return the Doppler shift from suspended solids and bubbles. The 316 stainless steel pressure transducer reads head above the sensor face, compensated for the dynamic pressure of the moving flow, to 10 m.

03 Changeover logic

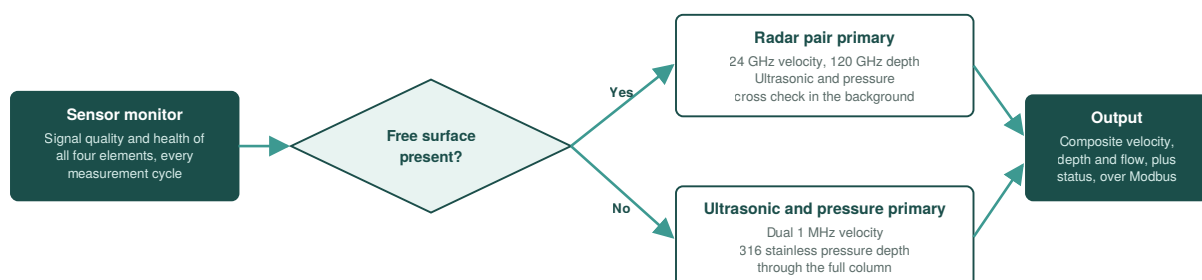


Figure 5. Selection of the primary sensor pair. Both pairs are monitored continuously; the transition into and out of surcharge is seamless with no blind zone.

04 Technical specifications

VELOCITY

Method	Dual mode Doppler: contactless 24 GHz radar in normal flow, dual 1 MHz contact ultrasonic in surcharge; seamless changeover, no blind zone
Range	0.050 to 20.000 m/s
Recommended operating condition	Above 0.2 m/s. Radar boundary region 0.05 to 0.2 m/s
Accuracy	Ultrasonic: 1 % of reading $\pm$ 0.005 m/s Radar: 2 % of reading $\pm$ 0.01 m/s
Resolution	0.001 m/s
Dead zone	Ultrasonic unidirectional 0 to 0.002 m/s Ultrasonic bidirectional $\pm$ 0.03 m/s Radar $\pm$ 0.05 m/s
Velocity radar	24 GHz continuous wave Doppler; antenna angle $24^\circ \times 12^\circ$; radiated power EIRP 20 dBm; antenna gain 18.5 dB
Ultrasonic transducers	Dual 1 MHz Doppler, 5° beam
Direction	Unidirectional model: absolute velocity in ultrasonic mode Bidirectional model: signed velocity and flow, reverse flow and backing up identified

DEPTH

Method	120 GHz FMCW radar in normal flow, sealed pressure transducer in surcharge; composite depth output with each value retained
Range	0 to 10 m
Accuracy	0.2 % of full scale $\pm$ 0.001 m
Resolution	0.001 m
Depth radar	120 GHz FMCW; measuring range 0.15 to 40 m, dependent on radar cross section; antenna angle $14^\circ \times 10^\circ$
Pressure transducer	316 stainless steel, sealed, high accuracy; velocity dependent dynamic pressure compensation built in

WATER TEMPERATURE

Range	0 to 60 °C
Accuracy	0.5 °C
Resolution	0.1 °C

ELECTRICAL AND COMMUNICATIONS

Supply voltage	8 to 15 VDC, 12 VDC nominal
Current	150 mA at 12 VDC
Interface	RS485, two wire
Protocol	Modbus RTU, slave. Register map supplied with the product
Output damping	Configurable
Response time	10 s
Firmware	Remote update over RS485 when paired with an EDS logger
Configuration	EMS-Flow commissioning software via EDS logger, from the surface

MECHANICAL AND ENVIRONMENTAL

Housing	Alloy and engineering plastic composite; corrosion, impact and water resistant
Sealing	Electronics fully potted; unaffected by temperature cycling
Wetted transducers	316 stainless steel pressure transducer; ultrasonic transducers sealed in the sensor face
Cable	Integral, length to suit maintenance hole depth, specified at order
Mounting	Pipe obvert on EDS mounting bracket, sensor face toward upstream; installed and retrieved from the maintenance hole opening
Fouling control	Obvert mounting keeps the body out of the flow path and prevents material build up on the sensor face
Service	Sewer maintenance holes, permanent or temporary; sewage, trade waste and combined flows

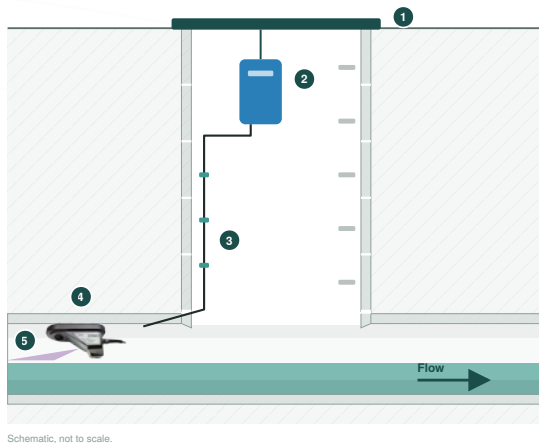
MODBUS PARAMETERS

Composite velocity	Mean velocity from the active sensor, m/s
Composite depth	Depth from the active sensor, m
Composite flow	Flow rate from velocity, depth and pipe profile
Radar velocity, radar depth	Individual values and status
Ultrasonic velocity, pressure depth	Individual values and status
Water temperature	°C
Active pair and sensor status	Which pair is in use; health of each element

Accuracy figures are stated under reference conditions. Radar velocity requires a free surface with sufficient roughness to return a signal; in the boundary region below 0.2 m/s the sensor prefers ultrasonic velocity where available. Specifications are subject to change without notice.

05 Installation arrangement

E-Flow 3 is fixed to the pipe obvert on an EDS mounting bracket with the sensor face toward the flow, and cabled to an EDS data logger suspended near the top of the maintenance hole. Bracket and cable are sized to pipe diameter and maintenance hole depth at order.

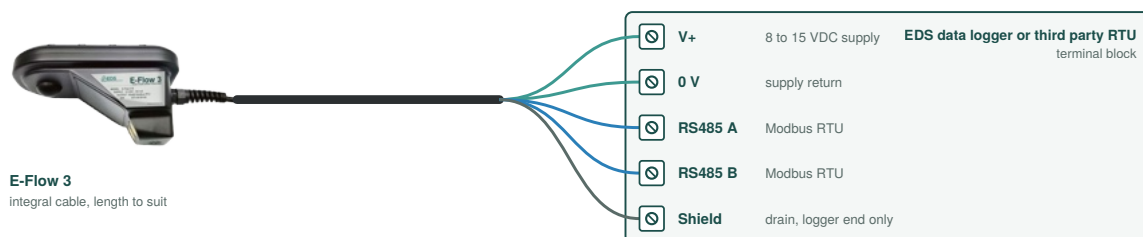


- 1 Cover and surface access**
Sensor installed and retrieved from the opening. Commissioned and verified from the surface with EMS-Flow, no confined space entry.
- 2 EDS data logger**
Suspended near the top of the shaft. Supplies 12 VDC, polls the sensor over RS485 and sends data to FlowSense over cellular telemetry.
- 3 Cable run**
Integral cable clipped to the shaft wall, length specified to suit maintenance hole depth.
- 4 E-Flow 3 at the obvert**
Bracket fixed to the pipe crown, sensor face toward the incoming flow. Body sits out of the flow path.
- 5 Measurement zone**
Radar pair to the free surface in normal flow; ultrasonic pair and pressure transducer through the column when full.

Figure 6. Typical maintenance hole arrangement.

06 Electrical interface

E-Flow 3 presents a single integral cable carrying supply and a two wire RS485 bus. It is polled as a Modbus RTU slave by an EDS data logger or any third party RTU, at 8 to 15 VDC and 150 mA.



Typical connection. Two wire RS485 with common supply return; core colours and termination are given in the installation guide supplied with the sensor.

Figure 7. Typical connection to a logger or RTU terminal block.

07 System architecture

In an EDS installation the sensor is polled by an EDS data logger in the maintenance hole, which forwards data over cellular telemetry to the FlowSense platform. Commissioning is done from the surface with EMS-Flow through the logger, without confined space entry.

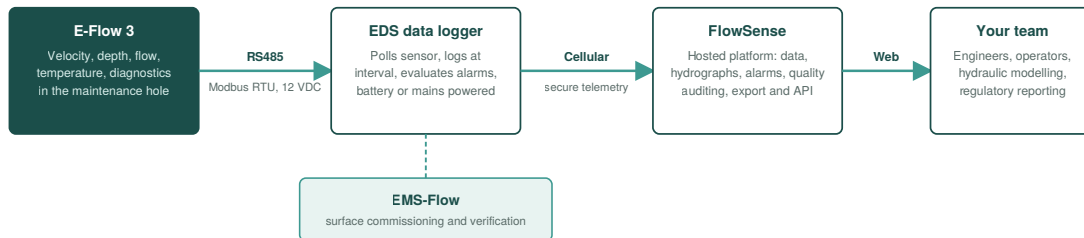


Figure 8. E-Flow 3 as part of the EDS monitoring system. The sensor can also be polled by third party RTUs and loggers over Modbus RTU.

Works with FlowSense

Hosted data, hydrographs, surcharge and reverse flow alarms, data quality auditing and export to hydraulic modelling tools. Per sensor diagnostics show which pair is active at any time, so a change in the record can be traced to a change in conditions rather than a fault.

EDS field crews install and commission across Australia, with rental fleets for temporary studies.

ORDERING

E-Flow 3 U	Unidirectional model. Ultrasonic mode reports absolute velocity.
E-Flow 3 B	Bidirectional model. Signed velocity and flow; identifies reverse flow and backing up.
Bracket	Obvert mounting bracket, sized to pipe diameter.
Cable	Integral cable, length to suit maintenance hole depth.
System	EDS data logger and telemetry, FlowSense subscription, installation by EDS field crews.

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